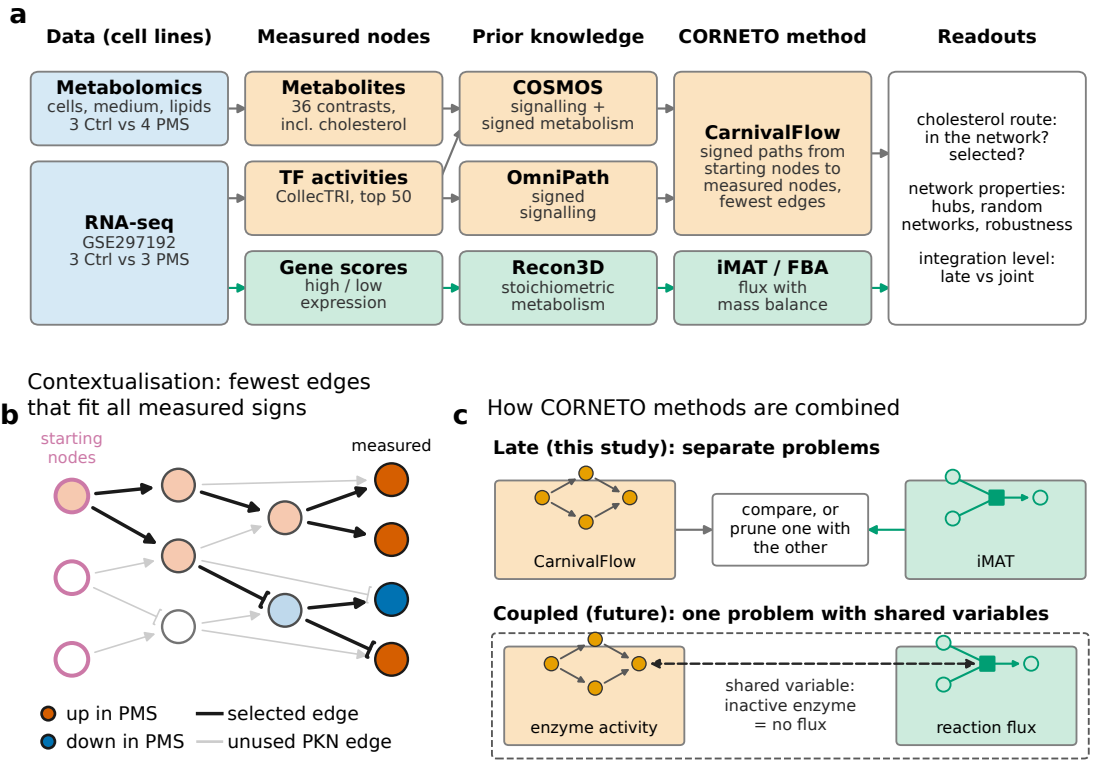
a
Data (cell lines)
Measured nodes
Prior knowledge
CORNETO method
Readouts
Metabolomics
cells, medium, lipids
3 Ctrl vs 4 PMS
RNA-seq
GSE297192
3 Ctrl vs 3 PMS
Metabolites
36 contrasts,
incl. cholesterol
TF activities
CollecTRI, top 50
Gene scores
high / low
expression
COSMOS
signalling +
signed metabolism
OmniPath
signed
signalling
Recon3D
stoichiometric
metabolism
CarnivalFlow
signed paths from
starting nodes to
measured nodes,
fewest edges
iMAT / FBA
flux with
mass balance
cholesterol route:
in the network?
selected?
network properties:
hubs, random
networks, robustness
integration level:
late vs joint
b
Contextualisation: fewest edges
that fit all measured signs
starting
nodes
measured
up in PMS
down in PMS
selected edge
unused PKN edge
c
How CORNETO methods are combined
Late (this study): separate problems
CarnivalFlow
compare, or
prune one with
the other
iMAT
Coupled (future): one problem with shared variables
enzyme activity
shared variable:
inactive enzyme
= no flux
reaction flux